

Reza M. Asiyabi

Based in **Edinburgh, UK** | Right to Work: **UK Global Talent Visa** (no sponsorship required in the UK)

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Summary

Geospatial AI/ML Research Scientist | Applied AI for Geospatial/Environmental Intelligence | Foundation Models

Machine learning and remote sensing scientist/engineer specializing in building scalable AI systems for large-scale Earth observation and geospatial data, with a focus on environmental and risk-related signals. Experienced in leading ML/EO projects at the University of Edinburgh, the UK's National Centre for Earth Observation (NCEO), Stanford University, and ESA-funded initiatives, developing models for multimodal satellite data (SAR, optical, LiDAR) and geospatial time series. Strong background in deep learning, representation learning, and physics-aware AI, with experience translating research into production-oriented pipelines. Skilled in Python and large-scale experimentation to deliver robust actionable insights from complex environmental data.

Core Technical Expertise and Skills

Geospatial AI & Earth Observation

- **Remote sensing data:** SAR, Multispectral, LiDAR, hyperspectral; sensor fusion and multi-modal geospatial analysis
- **EO processing tools:** Google Earth Engine, GDAL, Geopandas, Rasterio, Xarray, TerraTorch, TorchGeo, SNAP, ENVI, QGIS, ArcGIS
- **Application domains:** Environmental monitoring, land cover mapping, vegetation/crop health monitoring, change detection
- **Geospatial ML:** Geo-Foundation models, domain-aware and physics-informed learning, large-scale geospatial data pipelines

Machine Learning & Deep Learning

- **Frameworks:** PyTorch, JAX, TensorFlow/Keras, HuggingFace Transformers
- **Architectures:** CNNs, Vision Transformers, LSTMs, Autoencoders, Complex-Valued Networks, Concept Bottleneck Models
- **Specialisations:** Foundation models, self-supervised learning, multimodal fusion, few-shot learning, physics-informed neural networks, neural data compression, explainable AI (XAI)
- **MLOps:** End-to-end training pipelines (HPC & cloud), CI/CD, experiment tracking (TensorBoard, W&B), model deployment, version control (Git), proof-of-concept to production

Scientific Computing & Large-Scale Data

- **Python ecosystem:** NumPy, SciPy, Pandas, Scikit-Learn, OpenCV, Matplotlib
- **Large-scale pipelines:** Distributed data processing, cloud-based geospatial workflows, high-volume EO dataset management
- **Other:** AWS, R, MATLAB, SQL, C++ (working knowledge)

Research & Communication

- 20+ publications, 600+ citations; IEEE TGRS, J-STARS, J-STSP; cross-functional collaboration with academic and industry partners
- **Languages:** English (fluent), Persian/Farsi (native), Turkish/Azari (native)

Professional Experiences

Postdoctoral Research Associate (PDRA) – AI for Environmental Monitoring

Aug 2024 - Present

The University of Edinburgh – National Centre for Earth Observation (NCEO), Edinburgh, UK

Developed domain-aware AI systems for environmental monitoring using multimodal and noisy Earth Observation data

- Designed and implemented novel modular architectures (PG-CBM and StruMPL) for vegetation monitoring, integrating domain physics constraints with interpretable concept/multi-task learning to reduce prediction bias by 80% and improved ecological alignment.
- Built multi-stage learning pipeline with data augmentation and regularization strategies, combining multi-modal satellite and field data with physics-based loss functions to enforce structural constraints, improve model robustness against label noise, MNAR bias, domain shift, and spurious correlations in sparse-label regimes.
- Collaborated with domain scientists to design interpretable model outputs aligned with ecological theory.

Postdoctoral Research Assistant – Remote Sensing for Economical Monitoring

Jan 2024 to Jul 2024

Stanford University / Stockholm School of Economics, (Remote)

Remote work with the Center for Food Security and the Environment (FSE) at Stanford University on Market Activity Index (MAI) project.

- Designed novel remote sensing time-series change detection methods to detect and track economic activity of local markets.
- Implemented processing pipeline for satellite time-series imagery on Google Earth Engine and Google Cloud platform.

AI Research Engineer – Neural Data Compression

Jan 2022 to Jan 2023

European Space Agency (ESA) funded project - Adaptive SAR Signal Compression Through Artificial Intelligence (ARTISTE) Project

Led AI development for neural compression of raw SAR data for next-generation ESA satellite missions.

- Architected autoencoder-based compression pipeline using complex-valued neural networks to preserve phase coherency in SAR raw data. Achieved 8:1 compression ratio while maintaining signal fidelity for downstream processing.
- Designed custom loss functions balancing reconstruction accuracy with information-theoretic compression objectives.
- Implemented end-to-end training and evaluation framework and validated against ESA missions' operational SAR processing chains.
- Collaborated with engineering teams at DLR and Airbus to integrate ML compression module into mission simulation environment and assess computational feasibility for onboard processing.

Visiting Researcher – Complex-valued Deep Learning

Oct 2021 to Oct 2022

Zentrum für Sensorsysteme (ZESS), University of Siegen, Siegen, Germany

Research collaboration focused on theoretical and applied development of complex-valued neural networks.

- Designed novel complex-valued convolutional architectures preserving phase information in SAR imagery. Demonstrated ~20% accuracy improvement over real-valued baselines on classification tasks
- Developed theoretical framework for backpropagation and gradient computation in complex-valued networks, implemented custom PyTorch layers for complex activations and operations.

PhD Research Assistant – Deep Learning for Earth Observation

Dec 2020 to Dec 2023

CEOSpaceTech, University Politehnica Bucharest, Bucharest, Romania

Marie Skłodowska-Curie Early Stage Researcher (EU Horizon 2020 fellowship) conducting fundamental research on robust deep learning architectures for multimodal Earth Observation data.

- Developed end-to-end deep learning framework for complex-valued SAR data processing, combining semantic segmentation, classification, and neural compression in unified architectures.
- Designed coherency-preserving neural networks maintaining physical signal properties through complex-valued operations.
- Created and released S1SLC_CVDL: annotated complex-valued SAR dataset enabling reproducible research in complex-valued networks.

Selected Side Projects (More details on my GitHub)

- Deep Learning for BioMassters
 - Geospatial Foundation Model for AGB
- AGB Assistant LLM
 - EuroSAT Land Cover Classification
- Foundation Model Fine-Tuning
 - Geospatial Foundation Model Benchmark

Education

PhD of Electronics, Telecommunications and Information Technology

Dec 2020 to Dec 2023

University POLITEHNICA Bucharest (UPB), Romania

- Early-Stage Marie Skłodowska-Curie Early Stage Researcher, EU Horizon 2020 MENELAOS-NT ITN
- Dissertation: "Deep Learning for SAR Data in Presence of Adversarial Samples"

Master of Science (MSc) of Remote Sensing Engineering

Sep 2016 to Sep 2018

K.N. Toosi University of Technology, Tehran, Iran

- Thesis Title: Bag of Visual words Model enhancement for PolSAR Images Classification.

Bachelor of Science (BSc) of Geodesy and Geomatics

Sep 2012 to Sep 2016

K.N. Toosi University of Technology, Tehran, Iran

Fellowships, Awards, and Certificates

- Fellow of the University of Edinburgh's **Generative AI Laboratory (GAIL)** (2025-Present)
- Marie Skłodowska-Curie Doctoral Fellowship**, EU Horizon 2020 (2020-2023), MENELAOS-NT project, Grant No. 860370
- National Academic Excellence Fellowships**, K.N. Toosi University of Technology Top 1% nationwide university entrance exams (BSc 2012, MSc 2016)
- Certificate of the IEEE GRSS **High-Performance and Disruptive Computing** in Remote Sensing (2023)

Selected Publications (600+ citations on Google Scholar)

Selected Working Papers

- R. M. Asiyabi**, et al. " PG-CBM: Process-Guided Concept Bottleneck Models" (under review at IEEE TPAMI – Available on arXiv).
- R. M. Asiyabi**, et al. " StruMPL: Multi-task Dense Regression under Disjoint Partial Supervision and MNAR Labels" (under review at NeurIPS2026).

Selected Published Datasets

- R. M. Asiyabi**, et al., "S1SLC_CVDL: A Complex-Valued Annotated Single Look Complex Sentinel-1 SAR dataset for Complex-Valued Deep Networks", IEEE DataPort, (2023), doi: <https://dx.doi.org/10.21227/nm4g-yd98>.

Selected Journal and Conference Papers

- R. M. Asiyabi**, et al., "Complex-Valued Autoencoder-Based Neural Data Compression for SAR Raw Data," IEEE J-STSP, (2025).
- R. M. Asiyabi**, et al., "Complex-Valued End-to-End Deep Network with Coherency Preservation for Complex-Valued SAR Data Reconstruction and Classification," IEEE TGRS, (2023).
- R. M. Asiyabi**, et al., " Teaching AI to Think Like an Ecologist: Process-Guided Biomass Estimation from Earth Observation Data" *EurIPS – Workshop on AI for Climate and Conservation*, (2025).

Selected Presentations:

- Presentation at the AI/ML/EO sessions at the EurIPS 2025, ESA Living Planet Symposium 2022 & 2025, ESA Phi-lab 2021, IEEE IGARSS 2021 & 2022 & 2023 & 2024, EUSAR 2022 & 2024, CBMI 2023, and many more.